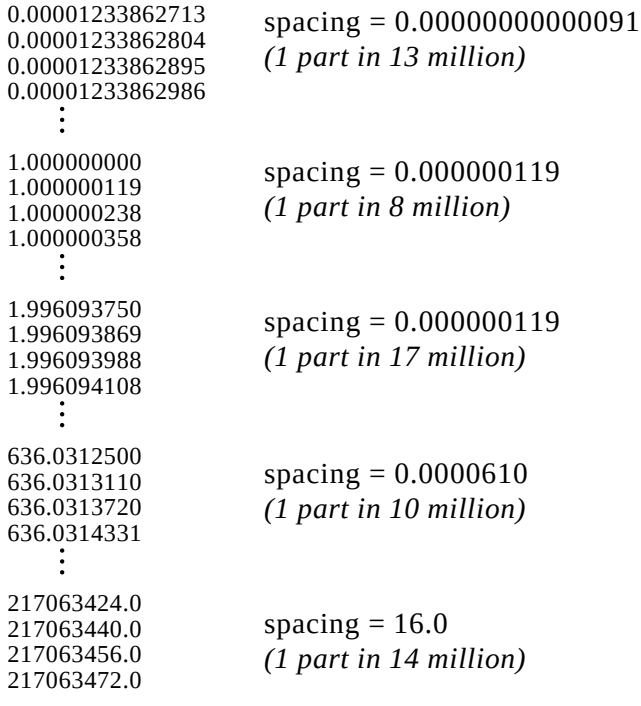

FIGURE 4-3
Examples of the spacing between single precision floating point numbers. The spacing between adjacent numbers is always between about 1 part in 8 million and 1 part in 17 million of the value of the number.

The program in Table 4-1 illustrates how **round-off error** (quantization error in math calculations) causes problems in DSP. Within the program loop, two random numbers are added to the floating point variable X, and then subtracted back out again. Ideally, this should do *nothing*. In reality, the round-off error from each of the arithmetic operations causes the value of X to gradually drift away from its initial value. This drift can take one of two forms depending on how the errors add together. If the round-off errors are randomly positive and negative, the value of the variable will randomly increase and decrease. If the errors are predominately of the same sign, the value of the variable will drift away much more rapidly and uniformly.

TABLE 4-1
Program for demonstrating floating point error accumulation. This program initially sets the value of X to 1.000000, and then runs through a loop that should ideally do *nothing*. During each loop, two random numbers, A and B, are added to X, and then subtracted back out. The accumulated error from these additions and subtraction causes X to wander from its initial value. As Fig. 4-4 shows, the error may be random or additive.

```
100  X = 1                'initialize X
110  '
120  FOR I% = 0 TO 2000
130    A = RND            'load random numbers
140    B = RND            'into A and B
150    '
160    X = X + A          'add A and B to X
170    X = X + B
180    X = X - A          'undo the additions
190    X = X - B
200    '
210    PRINT X            'ideally, X should be 1
220 NEXT I%
230 END
```